

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1700

Huntingdon District 138 kV
May 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1700. The Interconnection Customer proposes a 86.7 MW battery energy storage facility interconnecting at the Huntingdon District 138 kV substation in the Virginia area. The request was received on March 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 562 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,839,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Stonewall to Redbud Substation 138 KV Line
Contingency: Hp Hood Ct to Springfield 138 KV
Loading: 127.3% (Rating: 264 MVA)
DFAX: 5.2%
Pre-project loading: 125.6%
2. Facility: Santa Rosa Energy Ctr to Balch 138 KV L
Contingency: Hp Hood Ct to Springfield 138 KV
Loading: 122.5% (Rating: 318 MVA)
DFAX: 5.2%
Pre-project loading: 121.0%
3. Facility: 3 Mile Knoll Substation to Stonewall Substation 138 KV Line
Contingency: Springfield to Hp Hood Ct Substation 138 KV
Loading: 116.7% (Rating: 280 MVA)
DFAX: 5.2%
Pre-project loading: 115.0%
4. Facility: Wpcp Substation to Belleair 138 KV
Contingency: Santa Rosa Energy Ctr to Balch 138 KV
Loading: 115.7% (Rating: 278 MVA)
DFAX: 5.3%
Pre-project loading: 114.1%
5. Facility: Redbud to Balch 138 KV
Contingency: Springfield to Hp Hod Ct Substation 138 KV
Loading: 115.4% (Rating: 313 MVA)
DFAX: 5.2%
Pre-project loading: 114.0%

6. Facility: Old Chapel to Belleair Substation 138 KV
Contingency: Santa Rosa Energy Ctr to Balch 138 KV L
Loading: 111.2% (Rating: 309 MVA)
DFAX: 5.3%
Pre-project loading: 109.7%
7. Facility: Huntingdon District to Jet 138 KV
Contingency: Cleburne to Huntingdon District Substation 138 KV
Loading: 110.2% (Rating: 231 MVA)
DFAX: 60.8%
Pre-project loading: 87.4%
8. Facility: Springfield to Hp Hood Ct 138 KV L
Contingency: Santa Rosa Energy Ctr to Balch 138 KV
Loading: 107.1% (Rating: 348 MVA)
DFAX: 5.3%
Pre-project loading: 105.8%
9. Facility: BearIsland to Dunn Siding 345 KV
Contingency: Birchwood to Leavitt 345 KV
Loading: 104.9% (Rating: 2,769 MVA)
DFAX: 5.5%
Pre-project loading: 104.8%
10. Facility: Old Chapel to Hp Hood Ct 138 KV
Contingency: Balch to Santa Rosa Energy Ctr 138 KV line
Loading: 100.5% (Rating: 354 MVA)
DFAX: 5.3%
Pre-project loading: 99.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stonewall to Redbud Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$5,020,000	100% (\$5,020,000)
Santa Rosa Energy Ctr to Balch 138 KV	Rebuild line with higher-capacity conductor	\$4,962,000	100% (\$4,962,000)
Stonewall to 3 Mile Knoll 138 KV line	Rebuild line with higher-capacity conductor	\$3,804,000	100% (\$3,804,000)
Belleair to Wpcp 138 KV	Rebuild line with higher-capacity conductor	\$2,193,000	100% (\$2,193,000)
Redbud to Balch 138 KV	Rebuild line with higher-capacity conductor	\$2,801,000	100% (\$2,801,000)
Old Chapel to Belleair Substation 138 KV	Rebuild line with higher-capacity conductor	\$849,000	100% (\$849,000)
Jet to Huntingdon District 138 KV L	Rebuild line with higher-capacity conductor	\$7,995,000	39% (\$3,141,000)
Springfield to Hp Hood Ct 138 KV L	Reconductor line	\$1,777,000	100% (\$1,777,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bear Island to Dunn Siding 345 KV	Reconductor line	\$20,004,000	1% (\$142,000)
Hp Hood Ct to Old Chapel 138 KV	Reconductor line	\$8,150,000	100% (\$8,150,000)
	Total allocated to project		\$32,839,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.